Sinjalet çifte dhe tekë

Sinjali çift

$$x(-t) = x(t)$$

$$y(-n) = x(n)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

Sinjali tek

$$x(-t) = -x(t)$$

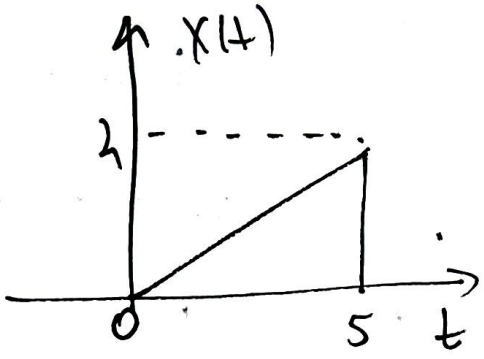
$$x(-n) = -x(n)$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

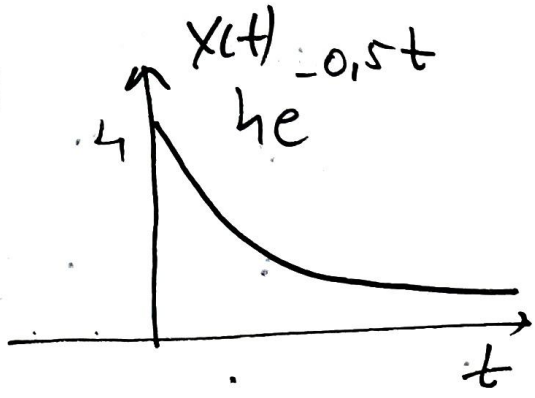
$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

1.5 Shkruani komponentet çifte dhe tekë të sinjaleve në figurë:

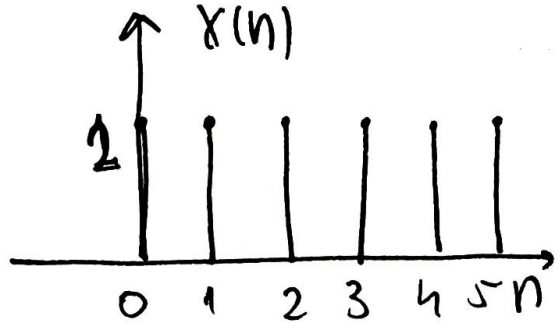
a)



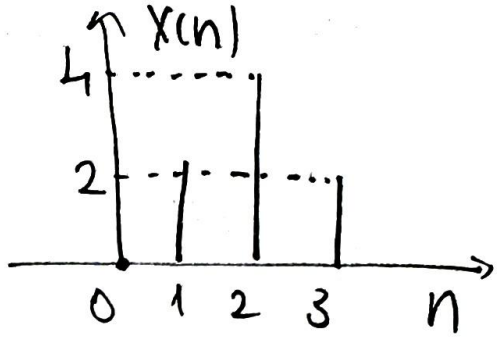
b)



c)

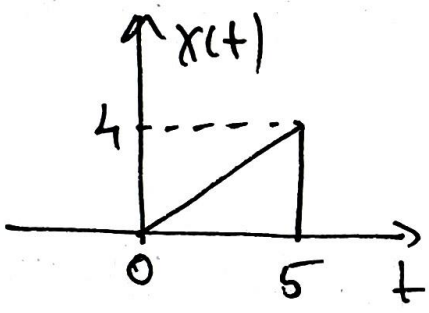


d)



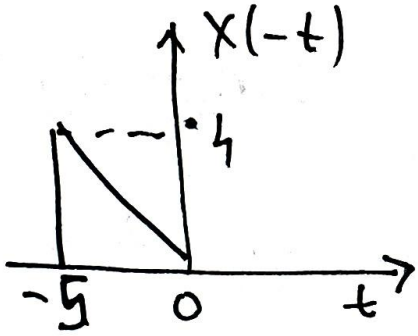
Zgjidhje

a)

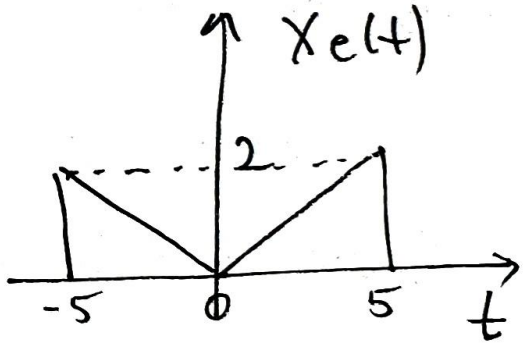


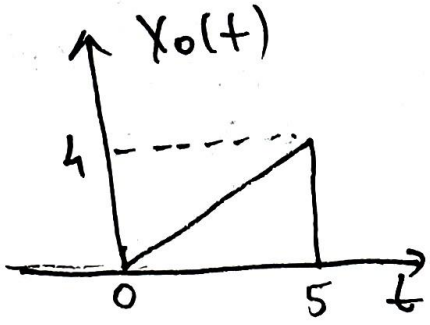
Pjesa çifte

$+$

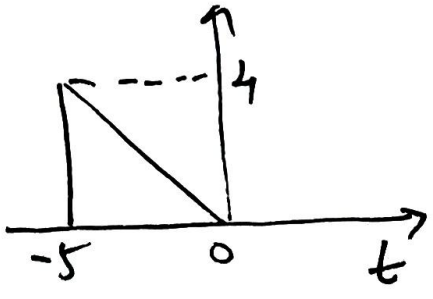


$\Rightarrow$

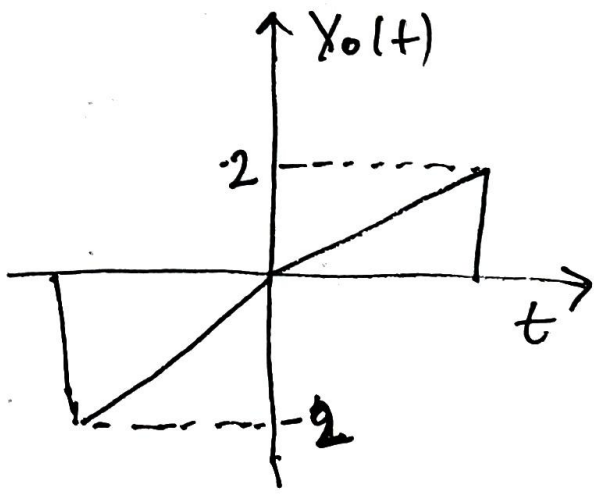




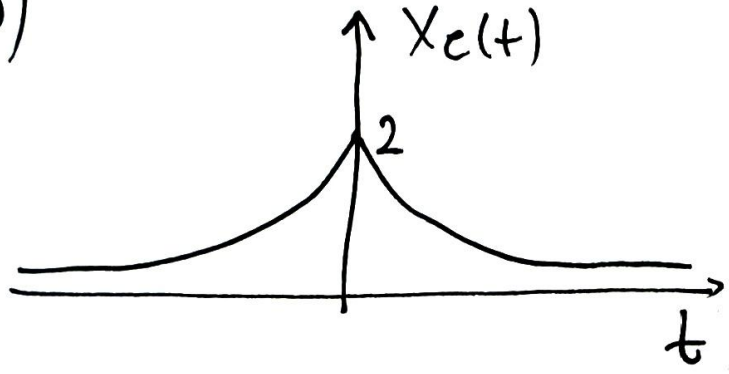
$-$

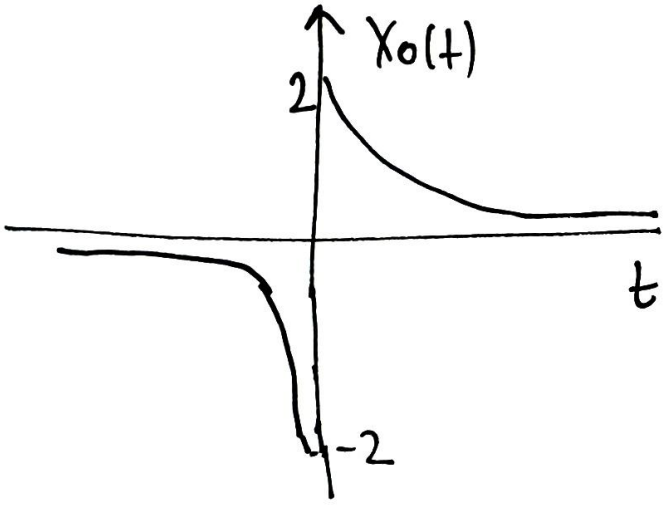


$\Rightarrow$

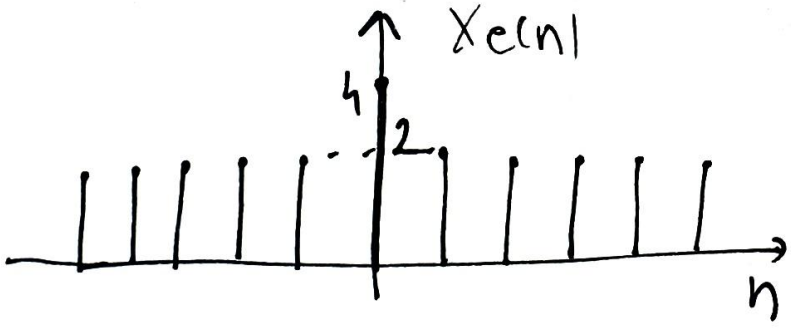


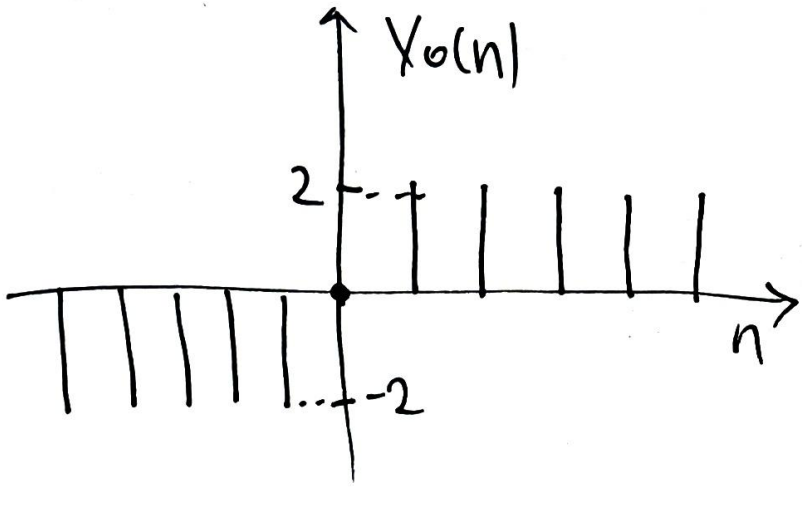
b)



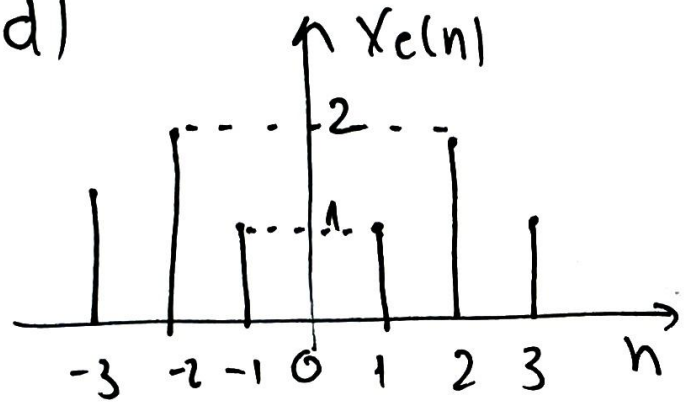


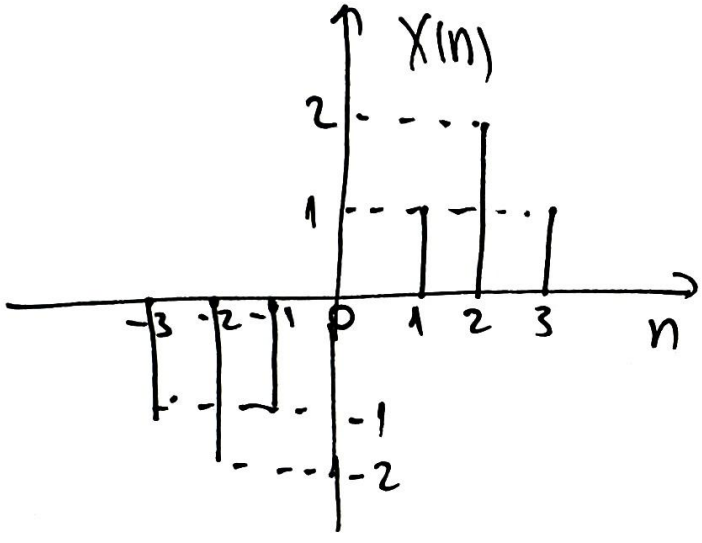
c)





d)





1.6 Gjeni pjesët çift dhe tek të sinjalit

$$X(t) = e^{jt} \quad \ldots\ldots\ldots\ldots$$

$$e^{jt} = X_e(t) + X_o(t)$$

$$X_e(t) = \frac{e^{jt} + e^{-jt}}{2} = \cos t$$

$$X_o(t) = \frac{e^{jt} - e^{-jt}}{2} = \sin t$$

} Formulat e Eulerit

Periodiciteti i sinjaleve

Sinjalet — periodike
Sinjalet — jopperiodike

* Sinjali i vazhdueshem $x(t)$ është periodik nëse mund të gjendet së paku një $T \in R$ për të cilen vlen:

$$x(t) = x(t+T)$$

* Sinjali diskret $x(n)$ është periodik nëse mund të gjendet së paku një numër i plotë $N \in Z$ për të cilen vlen:

$$x(n) = x[n+N]$$

1.10 $x(t) = \cos(\omega_0 t + \theta)$

vërteto që perioda e këtij sinjali është $\frac{2\pi}{\omega_0}$:

$$\cos(\omega_0(t+T)+\theta) = \cos(\omega_0 t + \theta + \omega_0 T)$$
$$= \cos(\omega_0 t + \theta)$$

Nëse $\omega_0 T = m 2\pi \Rightarrow \cdot T = m\frac{2\pi}{\omega_0} \rightarrow$ perioda

1.11. $x(n) = e^{j\Omega_0 n}$

Vërteto që sinjali është periodik vetëm nëse $\Omega_0/2\pi$ është numër racional:

$$e^{j\Omega_0(n+N)} = e^{j\Omega_0 n} \quad e^{j\Omega_0 N} = e^{j\Omega_0 n}$$

$$\Omega_0 N = m 2\pi$$

$$\frac{\Omega_0}{2\pi} = \frac{m}{N}$$

1.16

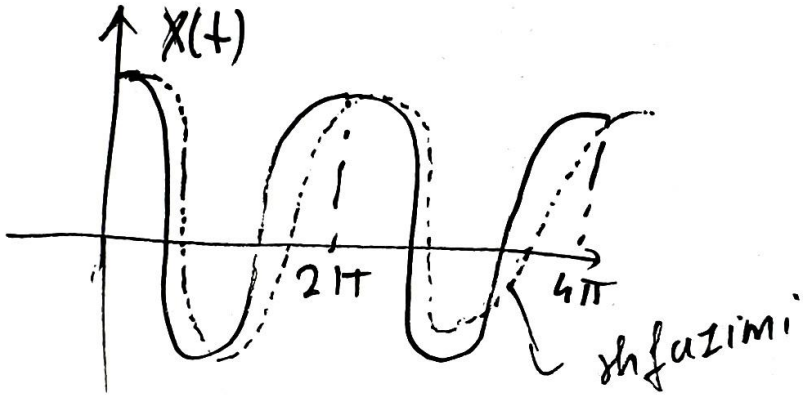
a) $x(t) = \cos\left(t + \frac{\pi}{4}\right)$

$\omega_0 = 1$

$\varphi = \frac{\pi}{4}$

$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{1} = 2\pi$

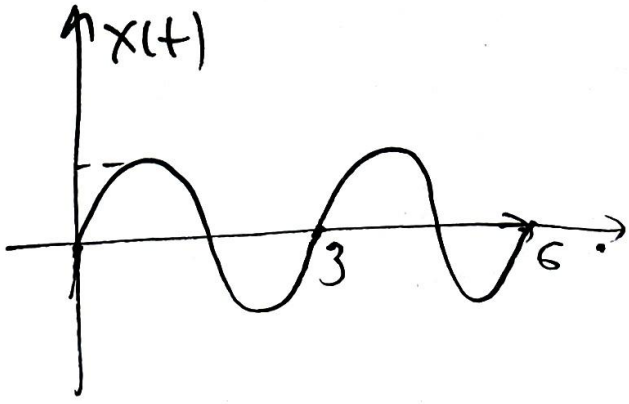
↓ sinjali periodik



b) $x(t) = \sin \frac{2\pi}{3} t$

$\omega_0 = \frac{2\pi}{3} \Rightarrow T = \frac{2\pi}{\omega_0} = \frac{2\pi}{\frac{2\pi}{3}} = \frac{6\pi}{2\pi} = 3$

↓ sinjali periodik



c)

$$x(t) = \cos\frac{\pi}{3}t + \sin\frac{\pi}{4}t = x_1(t) + x_2(t)$$

$$\omega_{01} = \frac{\pi}{3} \Rightarrow \quad T_1 = \frac{2\pi}{\frac{\pi}{3}} = 6$$

$$\omega_{02} = \frac{\pi}{4} \Rightarrow \quad T_2 = \frac{2\pi}{\frac{\pi}{4}} = 8$$

$$\frac{T_1}{T_2} = \frac{6^{:2}}{8^{:2}} = \frac{3}{4}$$

$$T_1 \cdot 4 = T_2 \cdot 3$$

$$6 \cdot 4 = 8 \cdot 3$$

$24 = 24$ → perioda themelore sinjali periodik

d)

$$x(t) = \cos t + \sin\sqrt{2}t = x_1(t) + x_2(t)$$

$$\omega_1 = 1 \Rightarrow \quad T = \frac{2\pi}{\omega_1} = 2\pi$$

$$\omega_2 = \sqrt{2} \Rightarrow \quad T_2 = \frac{2\pi}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\pi\sqrt{2}}{2} = \sqrt{2}\pi$$

$$\frac{T_1}{T_2} = \frac{2\pi}{\sqrt{2}\pi} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \sqrt{2} \rightarrow \text{sinjali jo periodik}$$

Discrete

a) $x(n) = \cos \frac{1}{4} n$

$\Omega_0 = \frac{1}{4}$

$\frac{\Omega_0}{2\pi} = \frac{\frac{1}{4}}{2\pi} = \frac{1}{8\pi}$ → singali l'aperiodik

b) $x(n) = \cos \frac{\pi}{3} n + \sin \frac{\pi}{4} n = x_1(n) + x_2(n)$

$x_1(n) = \cos \frac{\pi}{3} n \Rightarrow \Omega_1 = \frac{\pi}{3} \Rightarrow \frac{\frac{\pi}{3}}{2\pi} = \frac{1}{6}$

$\boxed{N_1 = 6}$

$x_2(n) = \sin \frac{\pi}{4} n \Rightarrow \Omega_2 = \frac{\pi}{4} \Rightarrow \frac{\frac{\pi}{4}}{2\pi} = \frac{1}{8}$

$\boxed{N_2 = 8}$

$\frac{N_1}{N_2} = \frac{6:2}{8:2} = \frac{3}{4}$

$N_1 \cdot 4 = N_2 \cdot 3$

$6 \cdot 4 = 8 \cdot 3$

$\boxed{24 = 24}$

Singali periodik ←

c) $x(n) = \cos^2 \frac{\pi}{8} n$

$$\boxed{\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)}$$

$$x(n) = \frac{1}{2}\left(1+\cos 2\frac{\pi}{8} n\right)$$

$$= \frac{1}{2}\left(1+\cos\frac{\pi}{4} n\right) = \frac{1}{2} + \frac{1}{2}\cos\frac{\pi}{4} n$$

$\downarrow x_1$ $\qquad$ $\downarrow x_2$

$$x_1(n) = \frac{1}{2} \Rightarrow \frac{1}{2}\cdot(1)^n = \frac{1}{2} \qquad \boxed{N_1 = 1}$$

$$x_2(n) = \frac{1}{2}\cos\frac{\pi}{4} n = \frac{1}{2}\cos \Omega_0 n$$

racional

$$\frac{\Omega_0}{2\pi} = \frac{\frac{\pi}{4}}{\frac{2\pi}{1}} = \frac{\pi}{8\pi} = \frac{1}{8} \rightarrow \qquad \boxed{N_2 = 8}$$